MILITARY SPECIFICATION

TEST STAND, AUTOMOTIVE GENERATOR, ALTERNATOR, STARTER, ASSOCIATED EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a 0- to 500-ampere capacity test stand for testing generators, alternators, starters, and their associated regulating and rectifying equipment utilized on tank and automotive equipment.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-C-88	- Casters, Rigid and Swivel, Industrial Duty
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-843	- Cushioning Material, Cellulosic
PPP-T-60	- Tape, Packaging, Waterproof

MILITARY

MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument)
MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	- Preservation-Packaging, Methods of
MIL-T-152	- Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	- Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronics, and Associated Equipment)
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank
MIL-M-3171	- Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity
MIL-S-3786	- Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification for
MIL-T-7928	- Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-B-22191	- Barrier Material, Transparent, Flexible, Heat Sealable
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for
MIL-I-24092	- Insulating Varnish, Electrical, Impregnating, Solvent Containing

STANDARDS

FEDERAL

FED-STD-595	- Colors
FED-STD-H28	- Screw-Thread Standards for Federal Services

MILITARY

AN4052	- Bushing-Generator and Power Take-Off Pad and Drive Adapter (for Converting to Type I or IA)
AND20002	- Drive-Type XII Engine Accessory
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-454	- Standard General Requirements for Electronic Equipment

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MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MIL-STD-1474 - Noise Limits for Army Materiel
MS3368 - Strap, Tiedown, Electrical Components, Identification, Adjustable, Self-Clinching, Plastic

* 2.1.2 Other Government documents. The following other Government documents, drawings, and publications form a part of this specification to the extend specified herein. Unless otherwise specified, the issues shall be those in effect on the day of the solicitation.

DOD STD-1866 - Soldering process, General (nonelectrical)
CID-AA-883 - Tape, Pressure Sensitive Adhesive, Masking
OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

DRAWINGS

ARMY

DETROIT ARSENAL

8699715 - Pulley, Generator
8699739 - Pulley, Generator
10884935 - Pulley, Alternator
10917484 - Pulley, Alternator

ROCK ISLAND ARSENAL

7550254 - Cable Assemblies

(Copies of specifications, standards, and other Government drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B633 - Electrodeposited Coating of Zinc on Iron and Steel
ASTM-D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

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MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C., 20037.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z55.1 - Gray Finishes for Industrial Apparatus and Equipment

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified (see 4.4 and 6.2.1), the contractor shall furnish one or more test stands for first article inspection.

3.2 Design. The test stand shall be designed for functional and conditional testing of 6-, 12-, and 24-volt generators, alternators, and their associated regulating and rectifying equipment as specified by the procuring activity (see 6.2.1). The test stand shall also incorporate provisions for free-running of 6-, 12-, and 24-volt starters as specified by the procuring activity (see 6.2.1). Test circuits shall include all necessary components and accessories required for a complete test setup. Controls, binding posts, and applicable connectors shall be oriented for optimum safety and ease of operation whether actuated by bare or gloved hands. The test stand shall be complete and when installed shall perform the operations specified herein.

3.2.1 Maintainability. The test stand shall be configured to reduce maintenance operations to the simplest procedures compatible with a minimum of maintenance skills and common equipment.

3.2.1.1 Service accessibility. The test stand shall be constructed so that disassembly and reassembly for maintenance service can be performed using only normally available standard tools and maintenance equipment. Accessibility shall be provided to permit servicing from the front or right-hand side of the stand.

3.2.2 Qualified products. Awards on these test stands will be made only when items covered by specifications requiring qualifications have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable qualified products list whether or not such products have been so listed by that time. The attention of suppliers is called to this requirement and manufacturers are urged to have such products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts for these test stands. Switches, terminals, meters, identification plates, and insulating varnish are covered by MIL-S-3786, MIL-T-7928, MIL-M-10304, MIL-P-19834, and

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MIL-I-24092, respectively.

3.2.3 Power supply. The test stand shall be designed to start and operate without malfunction of, or damage to, any component or part when supplied by any source of nominal 230/460-volt, 3-phase, 60-hertz (Hz) power.

For short time no-load starting (up to 2 minutes), the test stand shall start on voltage reduced by 50 percent, without damage to, or malfunction of, any component or part.

3.2.3.1 Relay excitation. The input power shall be reduced to provide 115 or less volts alternating current (Vac) or volts direct current (Vdc), as required, for operating test circuits.

3.2.4 Safety requirements. The test stand shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the test stand itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1).

3.2.4.1 Health and environmental requirements. Test stand design and construction shall incorporate noise limit requirements of MIL-STD-1474.

3.2.5 Electrical protection. The test stand circuits shall incorporate protection from damage in the event of internal failure resulting from high currents and from attempted operation on improper voltages. The main power inlet to the test stand shall be provided with overcurrent protection. Fuse holders and reset controls shall be readily accessible. All reset controls shall allow quick and easy resetting without the use of tools. If primary fuses are used, one spare set of three fuses of applicable type and value shall be provided. For all other fuses used within the test stand, five of each type and value used within the test stand shall be provided. Provision shall be made for storing the fuses in a readily accessible location within the test stand.

3.2.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.2.7 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerance in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.8 Threads. Threads shall conform to FED-STD-H28 and the applicable detailed standard section referenced therein.

3.2.9 Dimensions and weight. The test stand shall be designed within the following dimension and weight limits:

Height (maximum)	65 inches
Depth (maximum)	44 inches
Length (maximum, including all attachments and mounting brackets)	65 inches
Weight (maximum)	2,500 pounds

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3.3 Construction. All parts used in fabrication of the test stand shall be free of defects. In no event shall any process be used for reclaiming any parts of the test stand without the approval of the procuring activity. All surfaces of all parts shall be cleaned and free from extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness or adjustment. All parts subject to removal or adjustment shall not be swaged, peened, staked, or permanently deformed. Screw threads mating in aluminum or stainless steel shall be given an application of antiseize compound.

* 3.3.1 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the test stand industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. Soldering shall be in accordance with MIL-STD-454 Requirement 5.

3.3.2.1 Binding posts. Binding posts shall be of a size and composition to support connecting cables, and to carry continuously the maximum current of the circuit involved. Means shall be provided to preclude binding posts and connecting cables from becoming loose. Battery terminals on the test stand shall be provided with insulated knurled knobs for hand tightening of connections.

3.4 Performance and product characteristics.

3.4.1 Accuracy. Unless otherwise specified (see 6.2.1), test circuits of the test stand shall comply with requirements "a" through "d" of table I.

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TABLE I. Test circuit requirements

Conditions	Direct current (dc) voltage output test circuit	All other test circuits
	(Allowable indication error in percent of end scale value as defined in MIL-M-10304)	
a. Basic accuracy. Test stand stabilized at +70 degrees F to +90 degrees F.	±1 percent	±3 percent
b. Accuracy at high operating temperature. Test stand stabilized at +110 degrees F to +115 degrees F.	±2 percent	±3 percent
c. Accuracy at low operating temperature. Test stand stabilized at +30 degrees F to +35 degrees F.	±3 percent	±4 percent
d. Accuracy after all tests. Test stand being checked under same conditions as in "a" above.	Circuit indication shall not vary more than ±1 percent of end scale value from the readings obtained in "a" above.	

3.4.2 Environmental requirements. Unless otherwise specified (see 6.2.1), the test stand shall meet the environmental requirements of 3.4.2.1 through 3.4.2.3.

3.4.2.1 Storage temperatures. The test stand shall evidence no damage after having been:

a. Stabilized at an ambient air temperature of -60 degrees Fahrenheit (F) to -65 degrees F for a period of 72 hours.

b. Subjected to a 24-hour cycle of temperature, starting with 10 hours at +85 degrees F to +95 degrees F, followed by 5 hours of steady increase to +150 degrees F to +155 degrees F, 4 hours at +150 degrees F to +155 degrees F, and 5 hours of steady decrease to +70 degrees F to +90 degrees F.

3.4.2.2 Humidity conditions. The test stand shall evidence no damage after having been subjected to 20 hours exposure to, and while at, a relative humidity of 93 to 97 percent in an air temperature of +73 degrees F to +77 degrees F, followed by 4 hours exposure to, and while at, a relative humidity of 5 to 10 percent in an air temperature of +110 degrees F to +155 degrees F, air movement and radiation negligible.

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3.4.2.3 Vibration. The test stand shall evidence no damage after having been subjected to a vibration (simple harmonic motion) of 0.060 inch amplitude (total excursion 0.120 inch) through a varying frequency range in the vertical mode of 3 to 15 Hz, executed once per minute for a period of 1 hour when secured in an upright position utilizing the bolt holes in the base (see 3.5.22.1).

3.4.3 Dielectric strength. While in a relative humidity of 40 to 50 percent at an ambient air temperature of +70 degrees F to +90 degrees F, the drive motor, switches, controls, binding posts, and meters of the test stand shall withstand without damage, 1,000 volts root mean square (rms) at 60 Hz plus twice the rated voltage applied between mutually insulated circuits and between insulated circuits and ground.

3.5 Details of components.

* 3.5.1 Motor and drive unit. The motor and drive unit shall be shock mounted to absorb vibrations. Unless otherwise specified (see 6.2.1), the motor shall be a squirrel-cage induction type, design B, for operation on 230/460-volt, 60-Hz, 3-phase current in accordance with NEMA MG1. A variable speed drive shall be furnished to provide a speed range at the test generator or alternator drive shaft of 800 to 11,000 revolutions per minute (rpm). The full speed range shall be covered in not more than two ranges and with not more than two drive shafts. Drives shall be located on the right-hand side of the stand. The drives shall be infinitely variable throughout each range. If two drive shafts are used, they shall be located on the same end of the drive unit: one shaft shall provide a minimum speed range of 800 to 4,800 rpm; the other shaft shall provide a minimum speed range of 1,830 to 11,000 rpm. The height of the drive shaft or shafts shall be 27 inches ± 6 inches from the base of the test stand. A motor-generator set, if furnished, shall require only one electrical control to change speeds over the range from 800 to 11,000 rpm. If a two-speed motor is used, provision shall be made to prevent changing from one speed range to the other without stopping the drive motor. The power supplied to the test generator or alternator drive shaft shall be as specified in 3.5.1.2, with speed regulation and control as specified in 3.5.1.3 and 3.5.1.4. The drive shall allow operation for 144 hours continuously, under full-load, without damage to the motor or any part of the drive assembly. The complete drive assembly shall be securely grounded to the test stand cabinet.

3.5.1.1 Drive unit efficiency. With a 22-1/2 horsepower (hp) load on the drive shaft, and at any speed between 1,000 and 9,000 rpm, the overall efficiency of the drive and motor shall be not less than 75 percent.

3.5.1.2 Drive unit power. Minimum power available at the test generator or alternator drive shaft shall be as follows:

a. 4 hp at 800 rpm, linearly tapered to 22-1/2 hp at 1,000 rpm, continuous duty.

b. 22-1/2 hp at 1,000 to 9,000 rpm, continuous duty.

c. 22-1/2 hp at 9,000 rpm, linearly tapered to 4 hp, at 11,000 rpm, continuous duty.

d. 35 hp at 1,000 to 9,000 rpm, intermittent duty, 1 hour ON, 1 hour OFF.

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3.5.1.3 Drive unit fixed speed-load regulation. The test stand shall be designed to provide a fixed speed-load regulation of the generators or alternators under test. The regulated test generator or alternator drive shaft speed variance, for any fixed speed-load curve from no-load to full-load, shall not exceed 5 percent.

3.5.1.4 Drive unit speed control. The test stand drive speed control shall be designed to prevent surge or drift when changing test generator or alternator drive shaft speed from one speed setting to another speed setting at any given load from no-load to full-load. The speed control shall be located not less than 30 inches from the bottom of the test stand on the front panel. The control shall allow the operator to change speeds (within the full speed range of the test stand) in not more than 1 minute. When a speed control extension shaft is used, all supporting members shall be provided with bearings and lube fittings. Permanently lubricated bearings shall be considered acceptable. Bearings shall be of proper size and design to carry both the axial (thrust) and lateral loads imposed upon the speed control shaft. Speed setting shall be such that when changing from one set speed to a new set speed (within the speed range of the test stand, and at any load from no-load to full-load) the drive shaft shall conform to the following requirements:

a. Allowable speed variation shall be $\pm$1 percent.

b. The time required to reach a steady speed state after any speed change shall not exceed 3 seconds.

3.5.1.5 Drive motor connection board. The voltage connection links for the drive motor shall be located on a board oriented in such a manner as to allow the operator to alter the connections without exposing himself to a source of dangerous high voltage.

3.5.1.6 Drive motor starter. A magnetic-type starter with thermal overload protection shall be provided. The thermal overload shall open the power input circuit to the motor when the continuous load on the motor reaches 175 percent of rated load for 15 minutes. The starter control circuit shall be designed to permit starting the drive motor with input voltage reduced by 50 percent for a period of 2 minutes (during acceleration of the drive motor) without damage to any part. A cautionary plate shall be provided stipulating that the start button is to be depressed for not more than 30 seconds. Reversing of the drive motor shall be incorporated in the starter, or provided for by means of a reversing circuit of another type which incorporates protection against operation while the drive shaft is in motion. Starter control circuits shall be designed to operate on 115-volt, 60-Hz, single-phase current and shall be push-button operated. The control station shall be mounted on the front panel and shall be recessed to prevent accidental operation; push-buttons shall be color coded and nomenclatured to indicate their function. A push-button control shall also act as an emergency stop switch for the test stand drive motor. An indicating lamp shall be provided on the front panel to indicate when alternating current (ac) power is applied to the test stand. A second lamp shall indicate when the drive is in operation.

3.5.2 Load bank. The load bank shall consist of fixed and variable resistors with panel-mounted switches and shall be infinitely variable from no-load to 500-ampere load at 30 Vdc. In addition, provision shall be incorporated to permit selection of any load from no-load to 250 amperes at 15 Vdc. The load bank shall have a continuous duty rating, and shall be provided with a load disconnect circuit breaker-type switch that will allow repeated interruption of maximum load current, without damage to any part.

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3.5.2.1 Fixed-load resistors. Fixed-load resistors shall be so designed that, for any preset value of load between no-load and 500 amperes, the current shall not vary more than ±5 percent of the set value during 25 hours of continuous operation while the test stand is being subjected to any ambient air temperature from +30 degrees F to +115 degrees F. Load resistors having an inverse temperature coefficient are not acceptable.

3.5.2.2 Variable-load resistance. One variable-load resistor having a minimum rating of 25 amperes at 30 volts shall be provided. A variable-load resistance having an inverse temperature coefficient shall not be acceptable. The variable load bank control shall be equipped with an adjustment knob, or crank-type handwheel, not less than 4-1/2 inches nor more than 7-1/2 inches in diameter, possessing a mechanical advantage that will permit a force of not more than 40 inch-pounds (applied to the actuating device) to vary the load as required.

3.5.3 Tachometer circuit. A meter-indicating circuit shall be provided to indicate the test generator and alternator shaft speeds directly on the tachometer in two ranges. One range shall be from 0 to 5,000 rpm minimum to 6,000 rpm maximum. The second range shall be from 0 to 11,500 rpm minimum to 12,500 rpm maximum. Provisions shall be made to calibrate the tachometer for belt-driven generators and alternators so that the actual speed is indicated on the tachometer. The tachometer calibration shall cover generator and alternator pulleys ranging from 2 inches to 8 inches outside diameter. Calibration accuracy checks for pulley-driven generators and alternators will be based on pulley ratio effects only.

3.5.4 Output current, dc. A meter-indicating circuit shall be provided to indicate the dc output current of the test generators in three ranges: 15-0-50, 45-0-150, and 150-0-500 amperes.

3.5.5 Field current, dc. A meter-indicating circuit shall be provided to indicate the dc field current of the test generators and associated regulators in three ranges: 5-0-5, 15-0-15, and 30-0-30 amperes. (Generators shall be run self-excited for all tests.) Provisions shall be made for separately exciting the field on alternators.

3.5.6 Output voltage, dc. A meter-indicating circuit shall be provided to indicate the dc output voltage of the test generators in three ranges: 0-10, 0-20, and 0-50 volts.

3.5.7 Output current, ac. A meter-indicating circuit shall be provided to indicate line currents of 3-phase alternator outputs. The circuit shall be suitable for indicating currents from 0 to 500 amperes at frequencies varying from 70 to 800 Hz in two ranges: 0 to 100 amperes, and 0 to 500 amperes.

3.5.8 Output voltage, ac. A meter-indicating circuit shall be provided to indicate line voltages of 3-phase alternator outputs. The circuit shall be suitable for indicating voltages from 0 to 50 volts at frequencies varying from 70 to 800 Hz in two ranges: 0 to 25 volts, and 0 to 50 volts.

3.5.9 Millivolt drop circuit. A meter-indicating circuit shall be provided to indicate voltage drop across relay contact points. The circuit shall be suitable for indicating voltages in two ranges: 900-0-900 millivolts, and 9-0-9 volts. The circuit shall include a range selector switch and shall be controlled by a momentary "ON" switch. The circuit shall be protected in both scale ranges against damage from potentials higher than the maximum voltage indicated on the meter dial.

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3.5.10 Starting motor circuit. A meter-indicating circuit shall be provided for monitoring current when performing free-running tests on starting motors specified under 3.2. The circuit shall be suitable for indicating current in a range of 60-0-200 or 0-200 amperes for free-running tests. The circuit shall allow for making free-running tests with power from a vehicular generator driven by the variable speed drive. A voltage selector switch shall be provided and shall be of sufficient capacity for the performance of these tests.

3.5.11 Auxiliary circuits and controls.

3.5.11.1 Battery circuit. A battery circuit shall be provided to permit parallel connection of the 6-, 12-, and 24-volt terminals of the storage batteries (located within the test stand) with the resistor load bank. A circuit breaker shall be provided in the battery load circuit such that it shall not interfere with the generator output load tests or the starter tests. The circuit breaker shall be of such rating that the battery circuit involved shall be protected from damage due to battery current drain through the circuit when the generator speed is reduced.

3.5.11.2 Battery charge circuit. The test stand shall be provided with a constant current, manually operated battery charging circuit, for charging 6-, 12-, and 24-volt batteries, having an infinite current adjustment from 2.5 to 20 amperes. The circuit shall incorporate a 2-hour minimum timer. This power shall be available whether or not the drive unit is operating. A means shall be provided to indicate rate of charge.

3.5.11.3 Regulator test circuits. Regulator test circuits (both battery type and fixed resistor type), with panel-operated controls to select the load and type of load, shall be provided to test regulators specified under 3.2. (Regulators will be tested in conjunction with the generator or alternator and rectifier with which they are normally used in the field.) Regulator test circuits, in addition, shall incorporate provisions for performing tests on the generator control boxes specified under 3.2. A dc voltage circuit, independent of the batteries, variable from 0 to 32 volts, and designed to withstand a current of 15 amperes at 24 volts, shall be provided for testing the regulators specified under 3.2. This power shall be available whether or not the drive unit is operating.

3.5.11.4 Fixed resistors for regulators. High wattage resistors of 1/4 ohm, 1-1/2 ohm, 2-1/4 ohm, and 7 ohm shall be provided. The resistors shall be of the edge wound, ceramic or vitreous enamel type, and the value of resistance shall not change more than 12 percent when operating in any ambient air temperature from +30 degrees F to +115 degrees F.

3.5.11.5 Field rheostat. An infinitely variable (down to 0.10 amperes), step tapered, wire wound, continuous duty rheostat shall be connected internally to the field ammeter. The rheostat shall be suitable for controlling the fields of generators and alternators having field currents ranging from 0.2 to 30 amperes at voltages up to 30 Vdc. A control shall be provided for fine adjustment of the field current.

3.5.11.6 Voltage indicating circuit. A circuit, incorporating indicating lamps, shall be provided to indicate opening and closing of 6-, 12-, and 24-Vdc relays.

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3.5.12 Switches. All switches with knobs shall incorporate means to prevent the knobs from becoming loose.

3.5.12.1 Power input switch. The power input switch shall be located within the test stand cabinet and shall be accessible from the front of the stand for connections and maintenance. All ac power requirements for the test stand shall be taken from a single input. A suitable flexible conduit fitting shall be provided for the hole through which power input leads shall be installed. Locate this fitting in back panel.

3.5.12.2 Test circuit switches. Circuit switches shall be panel-mounted. All switches shall be capable of 10,000 cycles at the rate of 10 cycles per minute (cpm) while carrying rated current. The above provisions shall not apply to the starter control which shall conform to NEMA Standards.

3.5.12.3 Ammeter switch, dc. A switch conforming to MIL-S-3786 shall be provided to select ammeter ranges of 50, 150, and 500 amperes. It shall not be necessary to change test leads when changing load ranges.

3.5.12.4 Voltmeter switch, dc. A switch conforming to MIL-S-3786 shall be provided to select voltage ranges of 0-10, 0-20, and 0-50 volts.

3.5.12.5 Field current switch. A switch conforming to MIL-S-3786 shall be provided to select current ranges of 5, 15, and 30 amperes.

3.5.12.6 Polarity switch. A switch or switch activated contactors shall be provided for reversing ground polarity. The switch or contactors shall have a rating of not less than 500 amperes.

3.5.12.7 Load switches. A rotary switch or independent toggle switches shall be used to select load requirements. If a rotary switch is furnished, it shall function in connection with a magnetic contactor, or contactors, rated at not less than the maximum current of the circuit involved, and shall conform to MIL-S-3786. If independent toggle switches are furnished, they shall be rated at not less than the maximum current of the circuit involved.

3.5.12.8 Voltage selector switch. A switch, and magnetic contactors if necessary, shall be provided to select correct voltage and load resistances for regulators (see 3.5.11.4). Switches and magnetic contactors, if used, shall be rated at not less than the maximum current of the circuit involved.

3.5.12.9 Tachometer selector switch. A range selector shall be provided. The switch shall conform to MIL-S-3786.

3.5.12.10 Battery circuit switch. A selector switch in conjunction with indicating lamps shall be provided on the instrument panel to indicate the voltage that the batteries should be applying to the circuit. The switch selections shall provide for turning directly to "OFF" position from any voltage range (6-, 12-, or 24-volts). Indicating lamps shall have a service life of not less than 5,000 hours.

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3.5.12.11 Additional switches. Range selector switches shall be provided to select the required ranges in 3.5.7, 3.5.8, and 3.5.9 and phase selector switches for the required ranges in 3.5.7 and 3.5.8.

* 3.5.13 Leads and cables. The test stand shall be furnished with all leads and cables necessary for testing the equipment specified under 3.2. Leads, cables, and/or bundles shall be provided with identification tags in accordance with Military Standard MS3368, style optional, color natural. Test lead and cable terminals and corresponding terminals on the test stand shall be numbered or color coded to assure proper connection. For nonwaterproof generators and their associated regulators the other end shall be provided with insulated spade lugs to fit posts on the generators and their associated regulators specified under 3.2. Cables for connecting items specified under 3.2 to the test stand shall be furnished in accordance with drawing 7550254. When alligator clips are furnished on other test leads, suitable insulating boots must be furnished. Soldering of soldered terminals shall be in accordance with DOD-STD-1866. If crimp-type terminals are used, they shall be in accordance with type I, class 2 of MIL-T-7928. Sufficient battery leads and jumpers shall be furnished to provide complete connections for four 6-volt batteries (located within the test stand). Battery leads shall be terminated at each end by insulated connectors. Test leads and cables shall not be permanently attached to the test stand.

3.5.14 Calibration provisions. The test shall incorporate provisions to permit calibration of instrument circuits with external instruments.

3.5.15 Wiring. The internal wiring of the test stand shall be marked with an identifying letter and number. Each Alternating Current carrying wire shall be identified with the prefix letters "AC". Each Direct Current carrying wire shall be identified with the prefix letters "DC". In addition, each individual wire shall be marked with its own number. Wire length permitting, there shall be at least three marking imprints of identifying letter and number at two inch intervals from each end of the wire. Identification marking to be by a thermo-pressure contact process. The imprints shall not damage the wire. Wire and marking shall be of contrasting colors. Alternate method of wire marking must be approved by the Government. The wiring diagram shall reflect the test stand wire markings. Wire ends shall be fitted with plastic sleeves to reduce wear of insulation at terminals. Wiring harness shall be routed so as to preclude distortion, rupture, or other injury to the harness or clamps when opening and closing the main panel.

3.5.16 Meters. All meters of the test stand shall be of the 3-1/2 inch size, with exception of the dc ammeters which shall be of the 4-1/2 inch size. Meters shall be in accordance with MIL-M-10304.

3.5.16.1 Meter windows. Unless otherwise specified (see 6.2.1), meters shall be furnished with windows of glass or plastic material, free from discoloration, scratches, striae, electrostatic effect, or other defects which interfere in any way with normal reading of the dial scales. Plastic windows, when used, shall be in accordance with MIL-W-80, shall be a minimum of 3/32 inch thick, and shall be treated on both sides with an antistatic solution, silicon base preferred. Meter windows shall be secured by clamps or other suitable means (adhesive alone is not considered sufficient for holding windows in place).

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3.5.17 Generator and alternator mounting. Generator and alternator mounting shall be integral with the drive mounting and shall be as follows:

a. Generator and alternator mounting devices shall incorporate means for clamping all direct and pulley-driven generators and alternators specified under 3.2. A drive bushing conforming to AN4052 shall also be provided to convert the 1.2 PD Spline of the type XII drive (of AND20002). Mounting and clamping devices shall incorporate provisions to preclude distorting or deforming "soft" covers or shells of alternators.

b. If two drive shafts are used, mounting shall be so designed that the generator or alternator can be changed from one shaft to the other without removing clamps which secure the generator or alternator to the mounting. Provisions shall be made for accurately adjusting belt tension for pulley-driven generators and alternators. Tension adjustment and mounting means shall be designed to provide freedom from vibration.

3.5.18 Regulator and generator control box mountings. The test stand shall incorporate provisions for a readily mountable and demountable fixture for mounting the regulators and generator control boxes specified. The fixture shall mount to the right hand vertical column (structural member), when the test stand is viewed from the control and meter panel (front). The control box mounting surface shall be adjustable from 30 to 40 inches from the test stand base (floor). The mounting surface shall be adjustable through two axes to accommodate holding regulators and control boxes in their normal on-vehicle position. A spring loaded arm or other holding device shall be provided to hold the regulator covers in place, without the use of cover screws.

3.5.19 Rectifier mounting and cooling chamber. Provisions shall be made for mounting and cooling rectifiers specified under 3.2. The mounting shall be so designed that any of the rectifiers can be replaced by another rectifier of the same capacity. The cooling chamber shall be large enough to mount the rectifiers specified by the procuring activity. A minimum air flow of 1,000 cubic feet per minute shall pass through the plates of the rectifier. The cooling chamber shall be located inside the cabinet and shall be accessible from the front of the stand.

3.5.20 Drive adapters and belts. All adapters and belts necessary to drive the generators and alternators specified under 3.2 shall be furnished. One spare set of each type belt shall be provided. If multiple driven belts are used, the spare belts shall be supplied in matched sets. The number of belts in a set is determined by the maximum number of grooves on the driven pulleys. Pulleys required as adapters shall consist of a driving pulley for each output shaft. Three adapter pulleys with outside diameters of 4 inches shall be furnished. One pulley shall have four A-type grooves, one shall have four B-type grooves and one shall have four O-type grooves. Each pulley shall be marked with its type and groove angle. The following Army drawings are typical pulley configurations:

8699715	Pulley, Generator
8699739	Pulley, Generator
10884935	Pulley, Alternator
10917484	Pulley, Alternator

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3.5.21 Starting motor provision. Provisions shall be made, and equipment furnished with the test stand, for making free-running tests of the starting motors specified under 3.2. In addition, provisions shall be made for making free-running tests of the starting motors specified under 3.2 with power from a vehicular generator driven by the variable speed drive. A switch shall be provided whereby the operator, at his option, may disconnect the main drive motor to allow starter testing without operating the main drive. A hand tachometer shall be provided for use in conjunction with free-running starter tests. Changeover to starter test circuit shall be accomplished with reconnection of battery cables.

3.5.22 Cabinet. The test stand cabinet shall be made of sheet steel not less than Manufacturers' Standard Gage No. 16. The front panel shall have rounded edges. The top, side, and back panels of the test stand shall be separately removable without requiring complete disassembly of test stand. The cabinet shall house the drive assembly and shall be dimensionally in accordance with 3.2.9. A minimum work/storage area of 640 square inches shall be provided on the top rear area of the test stand. Edge molding around the work area of the cabinet shall be provided to prevent articles from falling off due to vibration during operation of the test stand. The work area shall be covered with insulating and scuff-resistant material (e.g., rubber mat).

3.5.22.1 Cabinet mounting. The test stand shall be skid-mounted and provided with four lifting eyes. All lifting eyes shall be removable to allow unobstructed use of the work area. Bolt holes shall be provided in the skid for securing the test stand to the bed of a truck van and so located that the bolts are completely accessible without any disassembly of the test stand. When specified (see 6.2.1), the test stands shall be mounted on four 6-inch casters conforming to type II, class 4, style 1 of FF-C-88. Openings shall be provided to allow the test stand to be transported or moved by fork lift. The size of the openings shall be 12 inches long by 3 inches high.

3.5.22.2 Battery compartment. The test stand cabinet shall provide a compartment large enough to house four storage batteries, each having dimensions of 11 inches in length, 8 inches in width, and 9-1/2 inches in height. Hold-downs and separators shall be provided to preclude damage to batteries or test stand. An exhaust vent shall be provided in the battery compartment. The vent shall be suited for connections with 1-inch flexible tubing. Flexible tubing shall be installed to prevent collapse due to bends or interference with other parts of the test stand. The vent tubing shall exhaust out the top panel of the cabinet. The battery compartment shall be sealed off from the drive assembly and shall have an interior coating of acid-resisting material to prevent corrosion. The compartment shall have an oak floor not less than 3/4 inch thick, coated with acid-resisting material. Provisions shall be made to permit checking the specific gravity of the battery electrolyte without disconnecting the battery cables, or removing batteries from the test stand (such as slide-out tray).

3.5.22.3 Storage compartment. A compartment with a hinged door shall be provided for storage of leads and spare parts furnished with the test stand. The compartment shall be accessible from the front or the side of the test stand on which equipment is tested. Leads and spare parts shall be secured to prevent damage to parts and test stand. Adequate space shall be provided to permit all storage compartment doors to be fully opened at any time, with or without the support brackets or other attachments in place, to provide necessary accessibility.

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3.5.22.4 High-voltage compartment. The power input switch and drive motor connection board shall be enclosed in a high-voltage compartment located in the front of the test stand cabinet. The cautionary phrase DANGER HIGH VOLTAGE shall be on the access door in red letters not less than 1 inch high. An interlock shall be provided to disconnect the drive motor while the access door is open.

3.5.22.5 Test stand cooling provisions. The test stand shall incorporate an air inlet and an exhaust manifold to provide cooling air for the test stand. The air inlet shall permit installation of a 6-inch diameter pipe to be fitted internally to the inlet. The exhaust manifold outlet shall permit installation of a 6-inch diameter pipe to be fitted externally to the outlet. The air inlet and exhaust outlet shall be located at the left side top panel of the test stand.

3.5.22.5.1 Load bank cooling. The load bank shall be fan cooled. The cooling unit shall provide a volume of air and air velocity that will remove enough heat from the load bank to preclude a temperature rise of more than 20 degrees F above ambient temperature at the surface of the load bank enclosure after 1 hour of operation with a 500-ampere, 30-volt load. Ambient temperature at the surface of the test stand prior to the 1-hour operation period shall be +70 degrees F to +90 degrees F. A protective device shall be provided to disconnect the input power to the test stand within 15 seconds if the cooling fan becomes inoperative.

3.5.22.6 Test panels. The test panels shall be made of the sheet steel not less than Manufacturers' Standard Gage No. 12. The panels shall be securely grounded. The meter instrument panel shall be mounted upright (vertical) or within 5 degrees to 15 degrees (canted rearward at the top) from vertical. A horizontal or horizontally sloped operator control panel shall be furnished. All instruments, controls, and terminals shall be mounted on the panels, and permanent and legible identification shall be marked on the panels or instruction plate to indicate their function. If etched metal nameplates are used, the background shall be painted to permit easy reading. A caution nameplate shall be attached to the panel above or below the tachometer. It shall have a red background with white lettering and shall read as follows:

CAUTION: DO NOT EXCEED MAXIMUM SPEED AS INDICATED IN TEST PROCEDURES.

All lettering on the test panels shall be not less than 1/8 inch in height.

3.5.22.7 Test panel arrangement. Terminals, binding posts, connectors, instruments, etc., necessary for testing regulators, control boxes, alternators, and generators shall be functionally grouped. Terminals for starter testing shall be located so as to have minimum influence on circuit and instrument readings. Each group shall be permanently outlined and marked to indicate its function. Where applicable, cable electrical connections on the test panel shall be provided with insulated knurled knobs for hand tightening of connections.

3.5.22.8 Test panel enclosure. The sides of the test panel enclosure shall be made of sheet steel not less than Manufacturers' Standard Gage No. 16. The back of the test panel enclosure shall be made of steel or expanded metal not less than the Manufacturers' Standard Gage No. 18.

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3.5.22.9 Test panel illumination. The test stand shall be equipped with a permanently mounted, adjustable, working lamp to illuminate the entire meter panel and working area. This lamp shall be placed so that it will not interfere with the operation of the test stand. It shall operate on 115 Vac, and shall be provided with an "ON-OFF" switch.

* 3.6 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Electromagnetic interference control. When specified (see 6.2.1), the test stand shall conform to the requirements of MIL-STD-461 for class IIB equipment.

3.8 Surface finishes.

3.8.1 Magnesium alloy parts. All exposed magnesium alloy parts shall be surface treated in accordance with MIL-M-3171.

* 3.8.2 Steel parts. All exposed steel parts, except bearing surfaces and those fabricated from oxidation-resisting steel, shall be zinc plated in accordance with ASTM B633. In cases where zinc plating is considered impractical or undesirable, the parts shall be painted.

3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing of the test stand shall be in accordance with the supplier's standard practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel or equivalent have been applied. The color shall be machinery gray according to ANSI Z55.1 (FED-STD-595) Chip 70.

3.10 Identification marking. The test stand shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.11 Information plates.

3.11.1 Identification plate. An identification plate, of material conforming to material requirements of MIL-P-514 or MIL-P-19834, permanently and legibly marked in accordance with 3.10 shall be securely attached to the outside of each test stand. All test stands of one manufacturer, supplied under a particular contract or purchase order, shall be consecutively serial-numbered.

3.11.2 Wiring diagram plate. An internal wiring diagram plate shall be provided on each test stand. It shall be of material conforming to material requirements of MIL-P-514 or MIL-P-19834, permanently and legibly marked and mounted in an accessible location. It shall show a complete wiring diagram covering all of the test circuits specified herein.

3.12 Workmanship. Workmanship shall be such as to assure equipment of a neat general appearance, uniform in quality, and free of defects that would adversely affect life or serviceability. Workmanship shall comply with MIL-STD-454, Requirement 9.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirement of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in applicable paragraphs of the specification.

4.4 First article inspection. First article inspection shall be applied to the preproduction sample or first production item noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5.2 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.5 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. For quality conformance inspection, a lot shall consist of all test stands of one manufacturer submitted for inspection at one time. Unless otherwise specified (see 6.2.1), the inspection shall consist of "a" through "h" as follows and failure of any unit to pass these inspections shall be cause for rejection.

a. Product examination (see 4.5.2).

b. Basic accuracy test (see 4.6.1).

c. Operational test (see 4.6.3).

d. Speed regulation test (see 4.6.7).

e. Speed setting test (see 4.6.8).

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f. Fixed load bank test (see 4.6.10).

g. Dielectric stress test (see 4.6.12).

h. Packaging inspection (see 4.7).

* 4.5.1 Sampling. Unless otherwise specified (see 6.2.1), smapling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following:

Inspection or Test	Inspection Level	AQL
Product examination (see 4.5.2)	100 percent inspection	
Basic accuracy test (see 4.6.1)	II	No failures allowed
Operational test (see 4.6.3)	II	No failures allowed
Speed regulation test (see 4.6.7)	II	No failures allowed
Speed setting test (see 4.6.8)	II	No failures allowed
Fixed load bank test (see 4.6.10)	II	No failures allowed
Dielectric stress test (see 4.6.12)	II	No failures allowed
Packaging inspection (see 4.7)	S-4	4.0

* 4.5.2 Product examination. Visually and manually examine the test stand to determine conformance with the requirements of 3.2 through 3.3.2.1, 3.5.1, 3.5.1.5 through 3.5.22.5, 3.5.22.6 through 3.6 and 3.8 through 3.12. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the test stand has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Basic accuracy test. Each test circuit shall be measured for ascending and descending readings at major scale graduations to determine compliance with the requirements of "a" in table I. Readings shall be compared with readings obtained on test instrumentation having an accuracy error of not more than 1/4 of 1 percent.

4.6.2 Switching tests. All switches, except starter control, that do not conform to MIL-S-3786 shall be subjected to an endurance test of 10,000 cycles at the rate of 10 cpm while carrying rated current. Each cycle shall consist of operating the switch through each of its positions and then back to the original position. Switches shall show no evidence of damage nor be inoperable following these tests (see 3.5.12.2).

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4.6.3 Operational test.

a. A 24-volt, 100- or 150-ampere alternator with matching regulator and rectifier shall be mounted and tested on the test stand in accordance with the operator's manual. Amperage, voltage, rpm, and field current draw shall be measured in accordance with the equipment manufacturer's specifications. Relay cut-out and relay contact voltage drop circuits shall also be tested (see 3.5.9 and 3.5.11.6).

b. Two starters, one of the jaw clutch type and one of the pinion type, shall be mounted on the test stand and subjected to free-running (no-load) tests (see 3.5.10). Testing voltage, amperage, and battery voltage shall be as recommended by the equipment manufacturer. This test shall be conducted by using the test leads, cables, and mounting adapters that are to be supplied with the test stand. The test stand shall indicate, within the tolerances specified in table I, the design characteristics of the equipment being tested.

4.6.4 Load bank cooling test. The test stand shall be operated for 1 hour with a 500-ampere, 30-volt load. Ambient temperature of the air surrounding the test stand shall have been stabilized for 4 hours at +70 degrees F to +90 degrees F prior to the 1 hour of operation. On conclusion of the 1 hour operating period, temperature readings shall be taken at not less than six points at the surface of the instrument panel and the surface of the panel directly over the load bank. Temperature pickups or thermometers utilized in this test shall have an accuracy within ± 2 degrees F. Temperature rise at any point shall not exceed 20 degrees F (see 3.5.22.5.1).

4.6.5 Endurance test. The test stand shall be subjected to the endurance tests in 4.6.5.1 and 4.6.5.2. Concurrent with the endurance test of the drive mechanism, a potential of 30 Vdc shall be applied to the load bank, and the resistance of the bank so adjusted that 500 amperes flow through the load bank. This potential shall be applied during the entire 144 hours of the continuous duty test. The load shall be applied directly to the test generator or alternator drive shaft. Tests shall be conducted in an ambient air temperature of +70 degrees F to +90 degrees F. At the conclusion of tests, the test stand shall show no evidence of damage to motor, bearings, gears, variable speed drive mechanism, or any other part of the drive assembly load bank or any component of the test stand (see 3.5.1).

* 4.6.5.1 Continuous-duty test.

Speed	Load	Time
800 rpm	4 hp	24 hours
1,000 rpm	22-1/2 hp	24 hours
2,000 rpm	22-1/2 hp	24 hours
4,000 rpm	22-1/2 hp	24 hours
9,000 rpm	22-1/2 hp	24 hours
11,000 rpm	4 hp	24 hours

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4.6.5.2 Intermittent-duty (1 hour ON, 1 hour OFF).

Speed	Load	Time
1,000 rpm	35 hp	24 hours
2,000 rpm	35 hp	24 hours
4,000 rpm	35 hp	24 hours
9,000 rpm	35 hp	24 hours

4.6.6 Horsepower output test.

* 4.6.6.1 Continuous duty. The hp output at the generator and alternator drive shafts shall be checked for 1 hour at each of the following speeds: 800, 1,000, 2,000, 4,000, 9,000 and 11,000 rpm. Output shall conform to the applicable requirements of 3.5.1.2.

4.6.6.2 Intermittent duty. The test stand shall be operated 1 hour ON and 1 hour OFF for two cycles at each of the following speeds: 1,000, 2,000, and 9,000 rpm. Output shall conform to the requirements of 3.5.1.2.

4.6.7 Speed regulation test. The speed regulation of the test stand drive unit shall be tested using a test method having an accuracy within 1/2 of 1 percent. The regulation shall be checked at the following speeds: 800, 2,000, 4,000, 6,000 and 9,000 rpm. Speed variation shall not exceed 5 percent (see 3.5.1.3).

4.6.8 Speed setting test. The speed control of the test stand drive unit shall be tested using a method having an accuracy within 1/4 of 1 percent. Speed setting shall be checked at no-load and full-load at 800, 2,000, 4,000, 6,000 and 9,000 rpm. Time required to change speeds shall conform to the requirements of 3.5.1.4.

4.6.9 Starter thermal overload test. The test shall be conducted in an ambient air temperature of +70 degrees F to +90 degrees F. With the test stand connected to a nominal 230-volt, 60-Hz, 3-phase source of power, and the generator drive shaft speed set at 1,000 rpm, a load shall be applied to the generator drive shaft. The load shall be increased until the thermal overload disconnects the power input to the test stand. The rate of load increase shall not exceed that specified in NEMA tests. The test shall be repeated at generator shaft speeds of 4,000 and 9,000 rpm. The percentage of overload shall be calculated to determine compliance with the requirements of 3.5.1.6.

4.6.10 Fixed load bank test. For first article tests, the test stand load bank shall be subjected to a 500 ampere load at a constant voltage of 24 Vdc for a period of 25 hours. During this period, the ambient air temperature shall be varied between 30 degrees F (or lower) and 115 degrees F (or higher), at a rate no greater than 10 degrees per hour. The current shall be monitored continuously during this test. For quality conformance inspection in 4.5 "f", the 500 ampere load at 24 Vdc shall be applied at room temperature and for 15 minutes only. Failure of either test, as evidenced by a variation in current of more than 5 percent from the initial value, shall be cause for rejection.

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4.6.11 Test circuit overload test. The test circuit shall be subject to momentary overload sufficient to assure that adequate protection devices are provided in accordance with 3.2.5.

4.6.12 Dielectric stress test. The test stand shall be subjected to 1,000 volts rms at 60 Hz plus twice rated voltage between mutually insulated circuits, and between ground and the following:

a. Power input line to motor.

b. Motor windings.

c. Binding posts.

d. Switch knobs.

e. Controls.

f. Meter Terminals.

g. Fuse disconnect switch.

Ambient temperature shall be stabilized at +70 degrees F to +90 degrees F, with relative humidity of 40 to 50 percent. Voltage shall be applied for a period of 60 seconds and shall be raised to its full value in steps not less than 500 volts per second and reduced by the same values. The accuracy of the measuring instrument shall be within 2 percent. After completion of test, the test stand shall show no evidence of damage (see 3.4.3).

4.6.13 Operating temperature test. The test stand shall be subjected to the high temperature condition "b" of table I until stabilized. While at that temperature, the test stand shall be tested as described in 4.6.1, except that accuracy shall be as required in "b" of table I. Operating temperature test shall be repeated under the low temperature condition "c" of table I.

4.6.14 Storage test. The test stand shall be subjected to the low temperature conditions of 3.4.2.1, then allowed to stabilize in an air temperature of +70 degrees F to +90 degrees F. The test stand shall then be inspected for evidence of damage and tested in accordance with 4.6.1 except that accuracy shall be as required in "d" of table I. This test shall be repeated after the test stand has been subjected to the high temperature conditions of 3.4.2.1.

4.6.15 Humidity test. The test stand shall be subjected to 20 hours exposure at a relative humidity of 93 to 97 percent in an air temperature of +73 degrees F to +77 degrees F, air movement and radiation negligible. The test stand shall then be subjected to a relative humidity of 5 to 10 percent in an air temperature of 110 to 115 degrees F, air movement and radiation negligible, for a period of 4 hours. At the end of this period, the test stand shall be tested for compliance with the accuracy test of 4.6.1. Failure of the test stand to meet the accuracy requirements shall be cause for rejection.

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4.6.16 Vibration test. The test stand shall be rigidly mounted on a vibration machine using the mounting bolt holes in the base (see 3.5.22.1). The test stand shall then be subjected to a vibration of simple harmonic motion of 0.060 inch amplitude (total excursion 0.120 inch) through a varying frequency range in the vertical mode of 3 to 15 Hz, executed once per minute, for a period of 1 hour in its normal mounting position. At the conclusion of this test, the test stand shall show no evidence of damage and shall perform within the accuracy requirements in "d" of table I when tested in accordance with 4.6.1 (see 3.4.2.3).

4.6.17 Electromagnetic interference control test. When specified (see 6.2.1), the test stand shall be subjected to all applicable tests required by MIL-STD-461 (see 3.7).

4.6.18 Battery charger test. The battery charging circuit shall be tested to verify compliance with the requirements of 3.5.11.2. Voltage selection, current adjustment, timer, and charging rate shall be as specified.

4.6.19 Equipment mounting and operational test. Tests shall be conducted on representative equipment as specified in the contract. The test stand shall provide for satisfactory mechanical mounting and electrical hook-up of the equipment, and shall indicate the design operation characteristics of the specified equipment to be tested.

4.7 Packaging inspection. Unless otherwise specified (see 6.2.1), inspection to determine compliance with packaging requirements of the applicable packaging documentation, for the degree of protection designated in the contract, shall be as specified in section 5.

5 PACKAGING

5.1 Military preservation. Preservation shall be as level A or level B, as specified (see 6.2.1).

5.1.1 Disassembly. The test stand shall be disassembled to effect savings in cubage and protect parts that are vulnerable to damage. A part shall not be removed from the assembly unless it can be reassembled readily in the field without special skills or special tools.

5.1.2 Level A.

5.1.2.1 Cleaning and drying. Components of the test stand shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.2.2 Unit packaging. The test stand shall be unit packed in accordance with method IIa of MIL-P-116. The instrument panel shall be covered with 1 inch thick uncompressed cushioning material conforming to PPP-C-843, type II. Corners of the cabinet and protruding parts, which are subject to puncturing or tearing the barrier bag, shall also be covered with cushioning material. The work light and battery cables shall be secured to restrict movement.

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5.1.2.2.1 Accessories and repair parts. Drive belts, hardware items, electric leads, and cables shall be packaged in bags fabricated from barrier material conforming to MIL-B-22191, Type III. Bag closure shall be by stapling or heat sealing. All remaining accessories shall be wrapped with similar type barrier material. Secure barrier materials with tape conforming to PPP-T-60. Place all accessories and repair parts in the storage compartment of the cabinet. Cushioning material conforming to PPP-C-843, type II shall be applied as necessary to restrict movement within the compartment.

5.1.2.2.2 Casters. When specified (see 3.5.22.1), casters shall be removed from the test stand and packaged in a fiberboard box conforming to PPP-B-636, class domestic. Place the casters in the storage or battery compartment.

5.1.2.2.3 Technical data. Publications issued with the test stand shall be packaged in accordance with method IC-1 of MIL-P-116. Attach publications to the outside of the method IIa package by utilizing pressure sensitive tape conforming to CID-A-A-883. The method IC-1 package shall be marked with the statement "PUBLICATIONS HERE" in 0.5-inch size letters.

5.1.3 Level B.

5.1.3.1 Cleaning and drying. Cleaning and drying shall be the same as for level A (see 5.1.2.1).

5.1.3.2 Unit packaging. The method of unit protection for the test stand shall conform to method IA-16 of MIL-P-116. The instrument gages shall be protected to guard against damage during shipment and storage. The work light and battery cables shall be secured to restrict movement.

5.1.3.2.1 Accessories and repair parts. Accessories shall be packaged the same as for level A (see 5.1.2.2.1).

5.1.3.2.2 Casters. When specified (see 3.5.22.1), casters shall be removed from the test stand and packaged in a paperboard box conforming to PPP-B-566. Place the casters in the storage or battery compartment.

5.1.3.2.3 Technical data. Technical data shall be packaged the same as for level A (see 5.1.2.2.3), except the publications shall be attached to the outside of the method IA-16 package.

5.2 Military packing. Packing shall be level A or level B, as specified (see 6.2.1).

5.2.1 Level A. Each test stand, packaged as specified in 5.1, shall be packed in a crate conforming to MIL-C-104, type II, class optional, style a. Arrangement of contents, anchoring, blocking, and bracing shall conform to MIL-STD-1186.

5.2.2 Level B. Each test stand, packaged as specified in 5.1, shall be packed in a crate conforming to MIL-C-3774, type I, style A. Anchoring of the stand to the skid base shall be in accordance with the appendix to the crate specification.

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* 5.3 Industrial packaging (preservation, packing and marking. When industrial packaging is specified (see 6.2.1), it shall conform to the requirements of ASTM-D3951.

5.4 Marking.

5.4.1 Military. When military levels of protection are specified, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and any special markings required by the contract.

6 NOTES

6.1 Intended use. This specification covers a test stand for use in testing the condition of tank and automotive 6-, 12-, and 24-volt generators, alternators, regulators, rectifiers, generator control boxes, and starters rated at 500-ampere capacity and less. These test stands are normally used in base maintenance shops, or in Shop Vans M-109 and M-220.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements:

a. Title, number and date of this specification.

b. When a first article is required for inspection and approval (see 3.1, 4.4, and 6.3).

c. The generators, alternators, and associated electrical equipment that the test stand shall be able to test (see 3.2).

d. The starters that the test stand shall be able to test (see 3.2).

e. Special safety requirements, if required (see 3.2.4).

f. Accuracy requirements, if different (see 3.4.1).

g. Environmental requirements, if different (see 3.4.2).

h. Motor, if different (see 3.5.1).

i. Meter windows, if different (see 3.5.16.1).

j. Caster mounting for cabinets, when required (see 3.5.22.1).

k. Electromagnetic interference control, when required (see 3.7).

l. Painting, if different (see 3.9).

m. Identification marking, if different (see 3.10).

n. Inspection conditions, if different (see 4.3).

o. First article inspection, if different (see 4.4).

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p. Quality conformance inspection, if different (see 4.5).

q. Sampling inspection and AQL(s), as applicable (see 4.5.1).

r. Electromagnetic interference control test, when required (see 4.6.17).

s. Packaging inspection, if different (see 4.7).

t. Selection of applicable degree of preservation-packaging (see 5.1, 5.2, and 5.3).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, assembly directions, and other instruction for operation, calibration, and maintenance, as identified on a DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample or a first production item as specified in 4.4. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Disposability methods. Environmental pollution preventative measures are contained in the materials specifications indicated herein. Refer to the materials specifications (or preparing activity) for recommended disposability methods.

6.5 Subject term (key word) listing.

Alternator
Dielectric Strength
Generator
Load bank
Rectifying
Relay excitation

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - AL
Air Force - 99

Review activities:
Army - AT

User activities:
Navy - MC

Preparing Activity:
Army - AL

Project No. 4910-0610

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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